



Quantifying noise complaint intensity via multi-LLM and revealing vertical greenery mitigation thresholds

Yan Zhang^a, Zhijie Zhang^b, Mei-Po Kwan^{c,a,*}, Wei Wang^d

^a Institute of Space and Earth Information Science, The Chinese University of Hong Kong, Pok Ying Tung Remote Sensing Science Building, Hong Kong Special Administrative Region of China

^b School of Geographic Sciences, Hebei Normal University, Shijiazhuang, Hebei, China

^c Department of Geography and Resource Management, The Chinese University of Hong Kong, Wong Foo Yuan Building, Hong Kong Special Administrative Region of China

^d State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, 430079, China

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ABSTRACT

Urban noise pollution has become a critical environmental issue affecting residents' quality of life. Existing research struggles to capture heterogeneity in noise sensitivity across different populations and cannot precisely quantify the mitigation effects of green space on noise. First, this study proposes an LLM-based method to quantify noise complaint intensity. Using four large language models, we constructed a benchmark dataset of noise complaint intensity. By replacing direct scoring tasks (where LLMs underperform) with comparison tasks (where LLMs excel), the fine-tuned RoBERTa model (a variant of BERT) achieved 61.48% accuracy when transferred to new datasets. This approach enables a fine-grained understanding of complaint data and captures large-scale distributions of resident complaint intensity. Second, green space exposure measurement integrated the NDVI from Sentinel-2 imagery and the 3D Green View Index (including tree and grass pixel ratios) from street view images. Generalized Additive Models results revealed significant vertical structural differences and non-linear threshold effects. The overhead greenspace coverage (NDVI) and grass coverage (low-level green space) showed highly similar trends with strong positive correlation: at low exposure levels, low-lying vegetation not only failed to mitigate noise but actually intensified complaint severity; whereas high-level green space (tall vertical greenery) represented by trees demonstrated strong noise suppression effects, though marginal benefits diminished at higher coverage (>0.12). In contrast, road density showed a reversal pattern of "initial suppression then amplification" (threshold ≈ 0.01). These findings indicate that constructing tall vertical green spaces exceeding effective thresholds is a crucial strategy for protecting vulnerable populations in urban planning.

1. Introduction

Noise pollution has become one of the most pervasive environmental stressors in contemporary urban environments (Welch et al., 2023; Song and Kwan, 2025; Tang and Li, 2024). As a key component of the "urban disease" syndrome, excessive noise exposure poses multifaceted threats to human health, including elevated cardiovascular disease risk, sleep disturbances, impaired cognitive function, psychological stress, and hearing damage (Poulsen et al., 2023; Thompson et al., 2022; Huang and Seto, 2024). The World Health Organization has identified environmental noise as a harmful stressor, with associated health costs reaching

billions of dollars annually (Kang et al., 2016; Hahad et al., 2025). Therefore, accurate measurement and assessment of noise pollution are crucial for building sustainable and inclusive cities that prioritize resident well-being (SDG) (Wang et al., 2024; Wang et al., 2025).

Current noise monitoring methods encompass a range of methodologies, including fixed monitoring stations, mobile acoustic measurements, noise mapping through modeling, satellite-based estimation, and crowdsourced data collection (Bello et al., 2019; Huang et al., 2024; Leach et al., 2024; Chen et al., 2025; Kumar and Kolhe, 2025; Zhang et al., 2026). Among these approaches, geotagged noise complaint data represents a unique and valuable source of information (Jiao et al.,

* Corresponding author. Institute of Space and Earth Information Science, The Chinese University of Hong Kong, Pok Ying Tung Remote Sensing Science Building, Hong Kong Special Administrative Region of China.

E-mail addresses: yanzhang@cuhk.edu.hk, sggzhang@whu.edu.cn (Y. Zhang), 20200813054@sjzc.edu.cn (Z. Zhang), mpk654@gmail.com, mpkwan@cuhk.edu.hk (M.-P. Kwan), wangwei8091@163.com (W. Wang).

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2025; Lu et al., 2025). Unlike purely physical measurements that only capture acoustic properties of sound, noise complaints reflect residents' subjective experiences and the social impacts of noise pollution. These complaints are direct expressions of noise hazards, documenting how noise disrupts daily life, interferes with essential activities, and undermines quality of life (Pinsonnault-Skvarenina et al., 2024). Recent studies have shown that AI-based and crowdsourced textual approaches can extract richer information from complaint or perception records, including BERT-based classification of noise complaints, random-forest prediction of repeat noise complaints (Zhang et al., 2024; Fan and Signorelli, 2024), and the identification of perceived thermal discomfort in underground metro systems (Chinazzo and Rotta Loria, 2026). More recently, LLM-based frameworks have been applied to uncover nonlinear spatial impacts on noise complaints and to classify complaint severity using probabilistic multi-agent reasoning (Lu et al., 2025; Chen et al., 2026; Yang et al., 2026). However, quantifying the impact of noise on urban systems based on geotagged complaint data primarily relies on analyzing the spatial distribution and density of complaints. For instance, Geographic Information Systems (GIS) are employed to map complaint hotspots and identify areas with high noise exposure levels (Song et al., 2026; Song et al., 2024; Chan et al., 2023). Other researchers have examined temporal patterns of noise complaints, revealing variations across different times of day, days of the week, and seasons (Lu et al., 2025; Zhang et al., 2026). Nevertheless, these studies predominantly treat each complaint as an equivalent unit, counting the number of complaints in specific areas or time periods without differentiating their severity.

This approach overlooks a critical dimension: heterogeneity in the perceived severity of noise disturbance expressed in complaint texts, which may partly reflect differences in noise sensitivity and vulnerability but cannot be directly interpreted as demographic group differences without individual-level demographic data (Wang et al., 2022; Zhang et al., 2026). The same construction noise from a building site may cause minimal disturbance to healthy adults but could severely affect vulnerable populations, including pregnant women, infants, elderly individuals, and patients with chronic diseases (Casey et al., 2017; Dzhambov and Lercher, 2019; Huang C H et al., 2024). Environmental health research demonstrates that noise sensitivity varies significantly based on demographic characteristics (Liu et al., 2023), health conditions, and psychosocial factors. For example, children exposed to chronic noise pollution exhibit impaired cognitive development and academic performance, while pregnant women experiencing high noise levels face increased risks of adverse birth outcomes (Bloemsma et al., 2019; Benocci et al., 2020). These vulnerable groups may suffer disproportionate impacts from noise exposure; yet, their voices may be drowned out in simple frequency/density-based analyses of complaints.

The inability to measure complaint intensity prevents researchers and policymakers from identifying the so-called “critical few”. Those sensitive individuals or communities suffering from extreme noise distress whose predicaments are masked by aggregate statistics. A neighborhood with relatively few complaints may be inhabited by residents experiencing severe noise impacts, while another area with numerous complaints may reflect only minor annoyances. Without distinguishing these scenarios, noise mitigation resources may be misallocated, failing to reach populations most in need of intervention. This limitation undermines efforts to achieve environmental justice in urban noise management.

Traditional sentiment analysis models offer potential solutions for assessing complaint intensity (Wu et al., 2025; Lian et al., 2026). Recent zero-shot LLM frameworks have further extended this line of research by enabling multimodal grievance classification and urgency scoring in civic feedback systems (Rajkumar et al., 2025). Based on natural language processing techniques, we can identify residents' emotional expression tendencies in noise complaints, such as whether they are positive or negative. However, these models suffer from several

significant limitations. First, generic sentiment analysis tools are not specifically designed for the noise complaint domain and cannot capture nuanced expressions and domain-specific terminology used by complainants. Second, these models typically produce simplified numerical scores (e.g., 0-1 scale). Since complaints represent concentrated expressions of resident dissatisfaction, sentiment analysis results are almost universally negative. It means that we cannot effectively quantify complaint intensity variations. The rapid development of Large Language Models (LLMs) provides new opportunities to address these challenges through their sophisticated natural language understanding capabilities and ability to capture contextual nuances (Zhang and Kwan, 2025; Zhang et al., 2022; Zhang et al., 2023).

Beyond the measurement challenges of noise complaint intensity, another critical knowledge gap exists regarding the potential buffering effects of green space exposure on noise complaint intensity. Although extensive research has established direct links between urban green exposure and noise pollution (Koprowska et al., 2018; Yu and Kwan, 2024), with green spaces providing multiple ecosystem services, including noise attenuation through vegetation sound absorption and psychological restorative functions enhancing stress recovery (Bloemsma et al., 2019; Ma et al., 2023). The relationship between green space exposure and noise complaint intensity (rather than complaint frequency) remains largely unexplored. We lack understanding of dose-response relationships: how much green space exposure is needed to effectively reduce noise complaint intensity? Does vertical vegetation structure matter in noise mitigation (Huang et al., 2025)? Furthermore, does this relationship exhibit threshold effects or diminishing marginal returns? Identifying and quantifying such threshold effects would be extremely useful for managing noise pollution and protecting vulnerable populations.

To address these two critical gaps, first, we developed a novel method using LLM-based ranking to measure noise complaint intensity. Since LLMs excel at comparison tasks rather than direct numerical scoring, we constructed a carefully annotated specialized noise complaint dataset and employed learning-to-rank techniques to establish a relative intensity scale. This approach enables us to capture subtle semantic differences in complaint texts and quantify the severity of noise impacts on populations. Second, leveraging this intensity measurement, we systematically examined the relationship between green space exposure and noise complaint intensity, charting dose-response curves and exploring how this relationship varies across different contexts and populations (Yu et al., 2024).

This study makes three key contributions. Methodologically, we demonstrate the potential of LLM-based approaches to extract meaningful insights from unstructured complaint data. We designed a replicable LLM-enhanced comparative ranking method applicable to noise exposure complaint research. Practically, our findings will provide evidence-based guidance for urban planners and policymakers, helping them implement precision-targeted noise mitigation strategies and nature-based solutions that prioritize the needs of vulnerable populations (Liu et al., 2025). In terms of application, we provide an LLM-annotated complaint intensity dataset applicable to noise complaints, advancing the application of AI in environmental health research.

2. Method

The research framework is illustrated in Fig. 1. It has two primary components: an LLM-Enhanced relative ranking model and dose-response curve calculation. First, we employed multiple large language models as “intelligent judges” to perform pairwise comparisons and rankings of complaint texts, constructing a standardized complaint intensity benchmark dataset. Through fine-tuning a RoBERTa model (a variant of BERT), we transferred the ranking knowledge to the full dataset, achieving scalable complaint intensity quantification. Subsequently, we integrated multi-source data (including satellite-derived

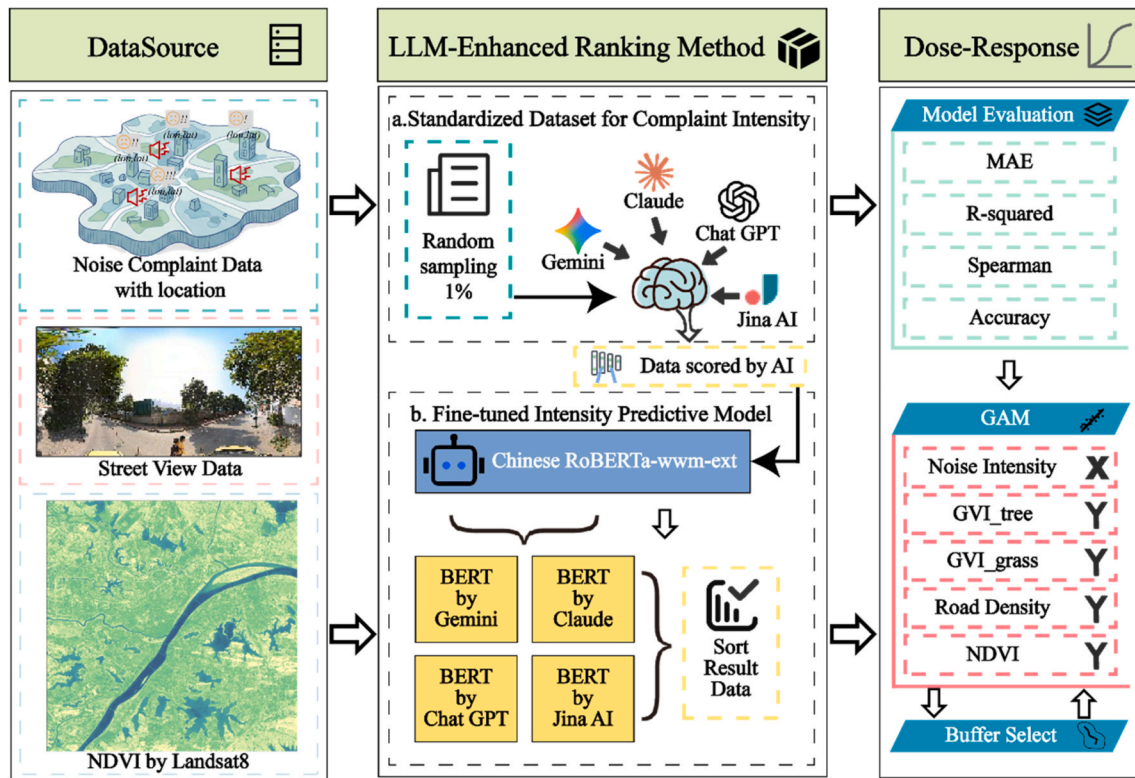


Fig. 1. Schematic diagram of the research framework.

NDVI, street view image (SVI) semantic segmentation, and road network density) and applied Generalized Additive Models to investigate the nonlinear mitigation effects of different types of green spaces (trees, grass) on noise complaint intensity, identifying critical dose-response thresholds to provide empirical evidence for nature-based noise mitigation strategies.

2.1. LLM-Enhanced Comparative Ranking Method for Noise Complaint Intensity

Traditional Natural Language Processing (NLP) methods often exhibit limitations when processing noise-complaint texts. Simple sentiment analysis can only determine the positive or negative polarity of text, struggling to distinguish the severity difference between “The music from the convenience store downstairs is a bit loud” (mild disturbance) and “Construction site working all night caused the entire family to be unable to sleep, child was frightened” (severe health hazard). Moreover, directly requesting models to score texts (pointwise scoring) typically faces calibration difficulties, as the model’s definition of “7 points” or “8 points” may fluctuate with context. In view of this, our study introduces large language models as “intelligent judges” (LLM-as-a-judge), drawing on the law of comparative judgment from psychometrics to construct absolute scales through relative comparison. It is one comparative quantification approach.

Fig. 2 illustrates the LLM-Enhanced comparative ranking method for noise complaint intensity. First, we employed stratified random sampling to select 270 complaints from the collected noise complaint dataset as a benchmark dataset. This dataset encompasses different types of noise sources (construction, traffic noise, commercial activities, neighborhood noise, etc.), different time periods (daytime, nighttime, early morning), and varying complaint expression intensities. Considering the advantages of large language models in pairwise comparison tasks, we adopted an iterative insertion sort strategy to rank these 270 complaints by severity. Specifically, we used four large language models, including Claude 3.5 Sonnet, GPT-4o, Gemini 2.0 Flash, and Jina, to

independently rank the benchmark dataset. For the Claude, GPT, and Gemini models, we employed an iterative insertion sort algorithm: initially placing the first complaint as the starting sorted list, then sequentially processing remaining complaints, with each API call determining where the new complaint should be inserted in the already-sorted list. The prompt was designed as follows:

“Your task is to determine where a new complaint should be inserted into the following sorted list. The entire list contains N complaints, ranked from most severe to least severe. Your evaluation criteria should only consider the noise intensity, impact scope, and duration reflected in the text content, rating those with particularly large scope, very long duration, and very high noise intensity as more severe. Please return only a number, which is the position index (from 0 to N) where the new complaint should be inserted. Extremely important: Your response must contain nothing other than the number—no text, punctuation, or explanation.”

This process requires 271 API calls to complete the full ranking of 270 data points. For the Jina Ranker model, we adopted a one-time batch reranking strategy: by defining a query statement representing “most severe complaint” characteristics (including keywords such as “noise, high decibels, nighttime construction, lasting months, vibration, unable to sleep, severe disturbance”), we submitted all 270 complaint texts at once to the API for semantic similarity calculation and ranking. This method requires only one reranking interface call to complete the entire dataset ranking. Ultimately, we obtained four independent ranking versions as different annotations of the benchmark dataset for subsequent model training and accuracy evaluation. To provide a human calibration reference, we additionally asked volunteers to score the benchmark complaints on a 1–10 severity scale, where 1 indicates the lowest perceived noise severity and 10 indicates the highest. The scores were normalized to 0–1 and used to train a Human reference model under the same RoBERTa training and evaluation framework.

To transfer the ranking knowledge from the benchmark dataset to large-scale unlabeled complaint data, we constructed a regression model

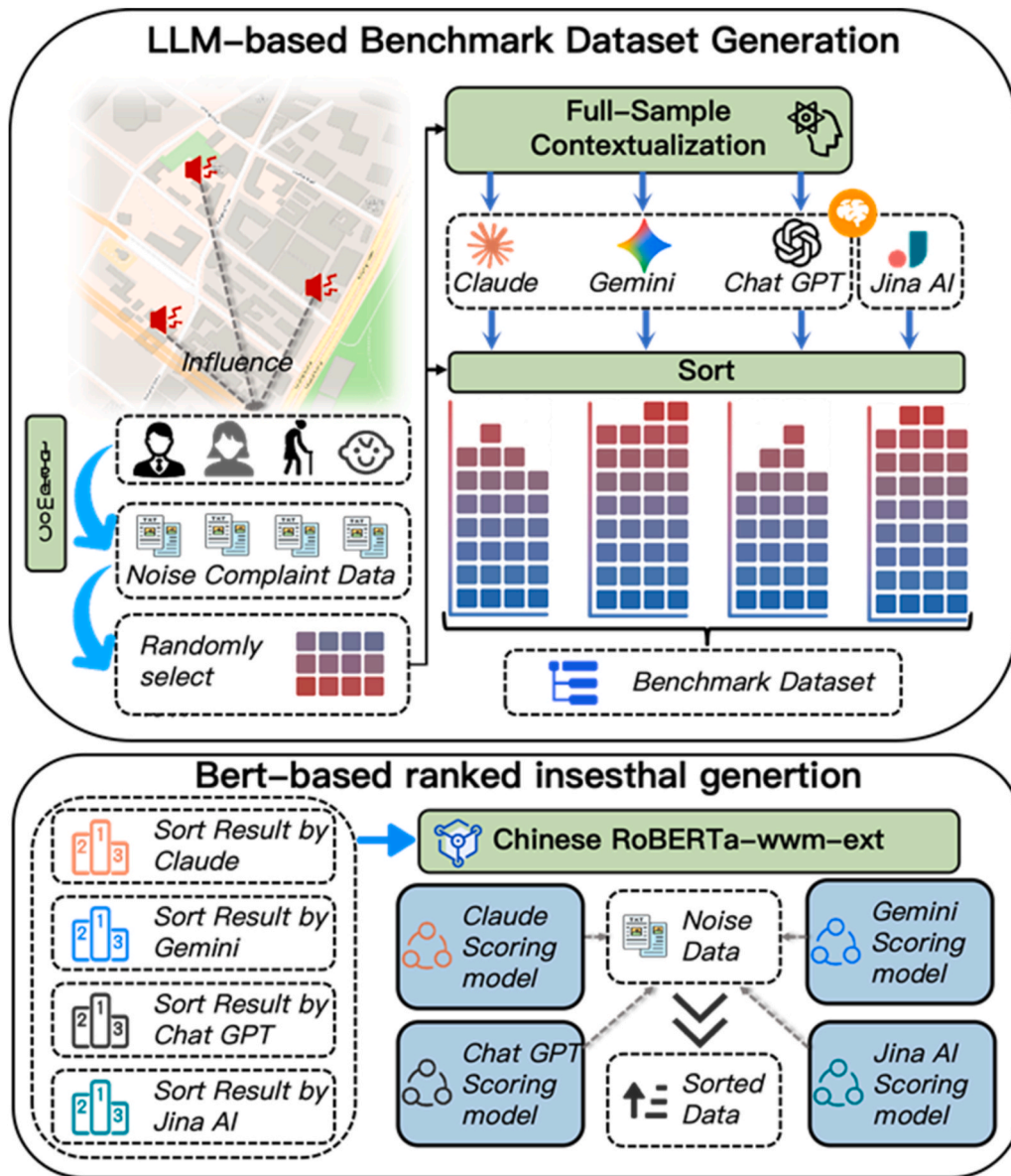


Fig. 2. The illustration of LLM-Enhanced Comparative Ranking Method for Noise Complaint Intensity (first ranking complaints based on different LLMs, then fine-tuning a BERT model to insert new complaint data into the benchmark dataset, with percentile ranking in the dataset representing complaint intensity).

based on Chinese RoBERTa-wwm-ext under a pre-training and fine-tuning paradigm (Cui et al., 2020), encompassing data preparation, model training, evaluation, prediction, and ranking generation. This model is pre-trained on large-scale Chinese corpora using whole-word masking and possesses robust Chinese semantic understanding capabilities. We converted severity rankings to continuous severity scores through min-max normalization:

$$\text{score} = (\text{max_rank} - \text{rank}) / (\text{max_rank} - \text{min_rank})$$

where higher scores indicate more severe complaints, rank ranges from 1 to 270, and scores range from [0, 1]. For model architecture, we retained RoBERTa's encoder layers and replaced the classification head with a single-output linear layer for regression tasks. Input texts were tokenized into token sequences with a maximum length of 128, and the model outputs continuous severity scores within the [0, 1] range. For the training strategy, we split the 270 annotated data into 70% training set and 30% validation set using stratified splitting to maintain ranking distribution, employed the AdamW optimizer with a learning rate set to 2×10^{-5} , batch size of 16, and used Mean Squared Error (MSE) as the

loss function: $L = (1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2$, where y_i is the true severity score and $\hat{y}_i$ is the model-predicted score. The model was trained for 15 epochs, with the best model selected based on Spearman's rank correlation coefficient on the validation set, and early stopping applied to prevent overfitting.

For the remaining unlabeled noise complaints, we used four trained BERT models (each trained on benchmark datasets annotated by Claude, GPT, Gemini, and Jina respectively) to generate severity scores (Some examples were shown in Table 1). During inference, the model outputs continuous scores in the [0, 1] range, which are then converted back to position values in the 1-270 range using the inverse normalization formula: $\text{position} = \text{max_rank} - \text{score} \cdot (\text{max_rank} - \text{min_rank})$. The Final rankings are assigned by sorting all complaints in descending order of predicted severity scores, where rank 1 represents the most severe complaint. This process generates four different versions of severity scores for each complaint, reflecting ranking differences under different LLM annotation strategies.

To evaluate the annotation quality of different large language models and the predictive performance of the BERT regression model, we

Table 1

Examples of model-predicted severity scores for unlabeled complaints.

ID	Longitude	Latitude	Sentiment	Predicted Severity	Predicted Position	Noise Complaint Content
1	114.3685	30.4653	0.5674	0.9407	12.8	Our home is 50 m from the railway ...
2	114.4017	30.4734	0.4534	0.9272	15.5	South of the community is the railway ...
3	114.4017	30.4734	0.4534	0.9272	15.5	South of the community is the railway ...
4	114.2176	30.6314	0.8608	0.9269	15.55	I live next to the Third Ring Road, trucks ...
5	114.1752	30.5966	0.798	0.9214	16.65	Every night dump trucks make extremely ...
...	...	...	...	...	...	...

employed four complementary metrics. The first metric is the Mean Absolute Error (MAE), which measures the average absolute difference between predicted and actual severity scores:

$$MAE = (1/n) \sum_{i=1}^n |y_i - \hat{y}_i|.$$

The second metric is the coefficient of determination R^2 , representing the proportion of variance in severity scores explained by the model:

$$R^2 = 1 - \left[\sum_{i=1}^n i = 1^n (y_i - \hat{y}_i)^2 \right] / \left[\sum_{i=1}^n i = 1^n (y_i - \bar{y})^2 \right]$$

where $\bar{y}$ is the mean of the true score. The third is Spearman's rank correlation coefficient, evaluating the monotonic relationship between the predicted and actual rankings, which is the primary metric for ordinal tasks:

$$\rho = 1 - \left[6 \sum_{i=1}^n d_i^2 \right] / [n(n^2 - 1)]$$

where d_i is the difference between the predicted and true rankings for the i -th sample. Finally, the accuracy within tolerance calculates the percentage of predictions falling within ± 0.1 of the true normalized score, assessing the model's practical acceptability. These metrics reflect model predictive capability from different perspectives, with Spearman's coefficient best capturing ranking quality and accuracy within tolerance, reflecting practical utility.

2.2. Urban green space exposure measurement

To assess the dose-response relationship between urban green spaces and noise perception, accurate measurement of green space is essential. Considering that different measurement approaches understand green space differently (Huang et al., 2025; Liu et al., 2026), we employed three green space exposure metrics simultaneously: the Normalized Difference Vegetation Index (NDVI), grass proportion in street view imagery (SVI), and tree proportion in SVI.

The NDVI quantifies vegetation density and health by comparing reflectance in the near-infrared (NIR) and red wavelength bands: $NDVI = (\rho_{NIR} - \rho_{Red}) / (\rho_{NIR} + \rho_{Red})$, where ρ_{NIR} represents near-infrared band reflectance and ρ_{Red} represents red band reflectance. NDVI values range from $[-1, 1]$, where negative values typically represent water bodies, values close to 0 represent bare soil or buildings, and positive values represent vegetation, with higher values indicating greater vegetation coverage and biomass. Considering that the NDVI primarily reflects the "quantity" of horizontal greening without distinguishing vertical vegetation structure, we further employed semantic segmentation technology to analyze SVI and calculate the three-dimensional Green View Index (GVI). The GVI comprises both vertical and horizontal greenery, represented by the proportion of trees and grass, respectively. We acquired panoramic images from major roads and residential areas in the study area through Baidu Street View API, with a sampling interval of 50 m, collecting approximately 15,000 SVIs. Using a pre-trained semantic segmentation model based on the DeepLabV3+ architecture (trained on Cityscapes and ADE20K datasets), we performed pixel-level classification of SVIs to identify green space elements such as

trees, grass, and shrubs. We calculated tree pixel proportion and grass pixel proportion separately as proxy variables for three-dimensional green space and planar green space (the latter representing lower-lying vegetation): $GVI_{tree} = N_{tree} / N_{total}$ and $GVI_{grass} = N_{grass} / N_{total}$, where N_{tree} and N_{grass} represent the number of pixels classified as trees and grass respectively, and N_{total} represents the total number of image pixels (Hou et al., 2025). Tree pixel proportion reflects the three-dimensional greening and canopy coverage provided by tall trees, while grass pixel proportion represents low-lying planar greening. These two indicators complement the NDVI to jointly depict the three-dimensional structural characteristics of urban green spaces (Zheng et al., 2024).

To quantify green space exposure levels surrounding complaint locations, we needed to determine the appropriate spatial buffer zone radii. Centered on each noise complaint point, we constructed buffer zones of 50m, 100m, 150m, 200m, 250m, and 300m respectively, extracting the mean NDVI, mean tree pixel proportion, and mean grass pixel proportion within each buffer zone as green space exposure variables. Simultaneously, we extracted the road network density within each buffer zone as an additional control variable to exclude the confounding effect of traffic density on noise complaint intensity. Road network density was calculated as the ratio of total road length within the buffer zone to buffer zone area:

$$\text{Road Density} = \frac{\sum L_{road}}{A_{buffer}}$$

where $\sum L_{road}$ represents the total length of all roads within the buffer zone, and A_{buffer} represents the buffer zone area.

2.3. Assessment of green space mitigation effects on noise complaint intensity

To capture the nonlinear relationships and potential threshold effects between green space exposure and noise complaint intensity, we employed GAM for dose-response analysis. GAM is a semi-parametric regression method that allows for nonlinear relationships between explanatory variables and response variables while maintaining model interpretability. It was selected as the primary analytical model because the objective of this study was explanatory rather than purely predictive. Specifically, we aimed to identify interpretable nonlinear exposure and response relationships between environmental factors and noise complaint intensity, and to further discuss potential threshold effects and planning implications. Compared with black-box machine learning models, GAM provides transparent partial response curves and allows spatial heterogeneity to be incorporated through smooth spatial terms.

The general form of the model is:

$$g(E[Y]) = \beta_0 + \sum_{j=1}^p f_j(X_j),$$

where Y is the response variable (noise complaint intensity), $g(\cdot)$ is the link function, β_0 is the intercept term, and $f_j(\cdot)$ is the smooth function for the j -th predictor variable X_j . In this study, we used an identity link function assuming complaint intensity follows a Gaussian distribution. The specific model form is:

$$Intensity = \beta_0 + f_1(NDVI) + f_2(GVI_{tree}) + f_3(GVI_{grass}) + f_4(Road_{Density}) + \varepsilon$$

where f_1 , f_2 , f_3 , and f_4 represent smooth functions for the NDVI, tree pixel proportion, grass pixel proportion, and road network density respectively, and ε is the error term.

We visualized the nonlinear relationships between green space exposure and noise complaint intensity by plotting partial effect curves for each predictor variable. The horizontal axis of the partial effect curve represents the range of the predictor variable values, while the vertical axis represents that variable's marginal contribution to the response variable. We calculated 95% confidence intervals through bootstrap resampling (represented by shaded bands around the curves), with horizontal density plots at the bottom showing the distribution density of complaint points across variable values. By observing changes in the slope of partial effect curves and intersections of confidence intervals, we can identify threshold effects of green space exposure. When green space indicators cross certain critical values, the direction or magnitude of their impact on noise complaint intensity changes significantly. This approach enables us to quantify the critical values where green spaces “take effect” and reach “saturation,” providing precise dosage guidance for urban greening planning.

3. Results

3.1. LLM-enhanced noise complaint intensity results

Our study area is Wuhan City, where we collected approximately 27,993 geotagged complaint text data points. And then we calculated the noise intensity (Fig. 3). This dataset represents a form of Volunteered Geographic Information (VGI). Additionally, we selected 270 noise complaints from this collection as our benchmark dataset, which we ranked by complaint intensity using various large language models. Since LLMs excel at comparison tasks rather than absolute scoring, we constructed a standard complaint intensity dataset to address this limitation. We can score the complaint intensity of unlabeled data based on its position within this dataset.

The BERT regression models trained on benchmark datasets annotated by four different large language models demonstrated excellent and comparable predictive performance on the validation set. Table 2 presents the performance of the four models across multiple evaluation metrics. The model trained on Jina annotations performed best across all metrics, with a Mean Absolute Error (MAE) of 0.1067, a coefficient of determination (R^2) of 0.754, a Spearman's rank correlation coefficient of 0.8789, and an accuracy of 61.48%. The Gemini-annotated model

Table 2

Performance comparison of complaint severity prediction models trained with different annotation sources.

Model	MAE↓	R-squared↑	Spearman ↑	Accuracy↑
Jina	0.1067	0.754	0.8789	61.48%
Gemini	0.1545	0.4846	0.7412	44.07%
Claude	0.1619	0.5279	0.7285	38.15%
GPT	0.1717	0.4601	0.7209	33.70%
Human	0.0838	0.7943	0.9015	71.11%

ranked second, with an MAE of 0.1545, an R^2 of 0.4846, a Spearman's rank correlation coefficient of 0.7412, and an accuracy of 44.07%. The model trained on GPT-annotated data showed relatively lower performance, with an MAE of 0.1717, an R^2 of 0.4601, a Spearman's rank correlation coefficient of 0.7209, and an accuracy of 33.70%. Overall, Jina and Gemini demonstrated superior performance. Table 2 shows that the human-based model achieved the best predictive performance, with an MAE of 0.0838, an R^2 of 0.7943, a Spearman correlation of 0.9015, and an accuracy of 71.11%. This confirms the value of human scoring as a calibration benchmark. AI can achieve accuracy close to human-level performance but at a significantly lower cost. However, human ratings tend to be more polarized, whereas AI demonstrates better consistency (Fig. 4).

Additionally, as also shown in Fig. 4 (Distribution of Predicted Severity Scores Across Different Models), which displays the distribution of predicted intensity for unlabeled complaint data, Jina exhibits a tendency toward normal distribution (indicating good consistency between prediction and training sets), while Claude and GPT tend toward lower scores (i.e., predicting lower complaint intensity), and Gemini produces relatively higher scores (i.e., predicting higher complaint intensity). The Human model shows a more polarized distribution, likely reflecting the limited number of volunteers and subjective differences in manual scoring. In contrast, the LLM-driven models produce smoother severity-score distributions, suggesting better scalability for large-scale annotation, although their accuracy is slightly lower than that of the Human model. Overall, our fine-tuned model successfully learned the ranking logic of large language models and maintained robust ranking capability on unseen complaint texts.

Specifically, low-intensity complaints (score <0.3) typically contain vague noise descriptions such as “a bit noisy” or “occasional noise,” with minimal mention of specific impacts on daily life. Medium-intensity complaints (score 0.3–0.6) begin to feature explicit noise source descriptions (such as “renovation” or “square dancing”) and mention basic

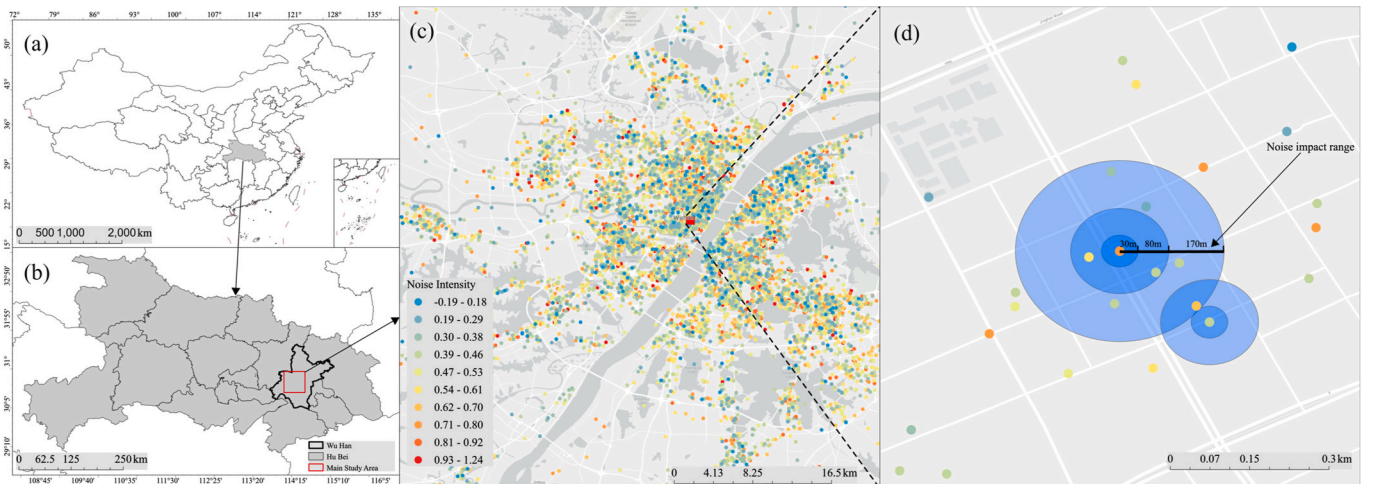


Fig. 3. The illustration of our study area and the distribution of calculated noise complaint intensity. (We constructed a buffer zone around each noise complaint location and collected information, including the NDVI at complaint locations, semantic segmentation of street view imagery, and road network density.)

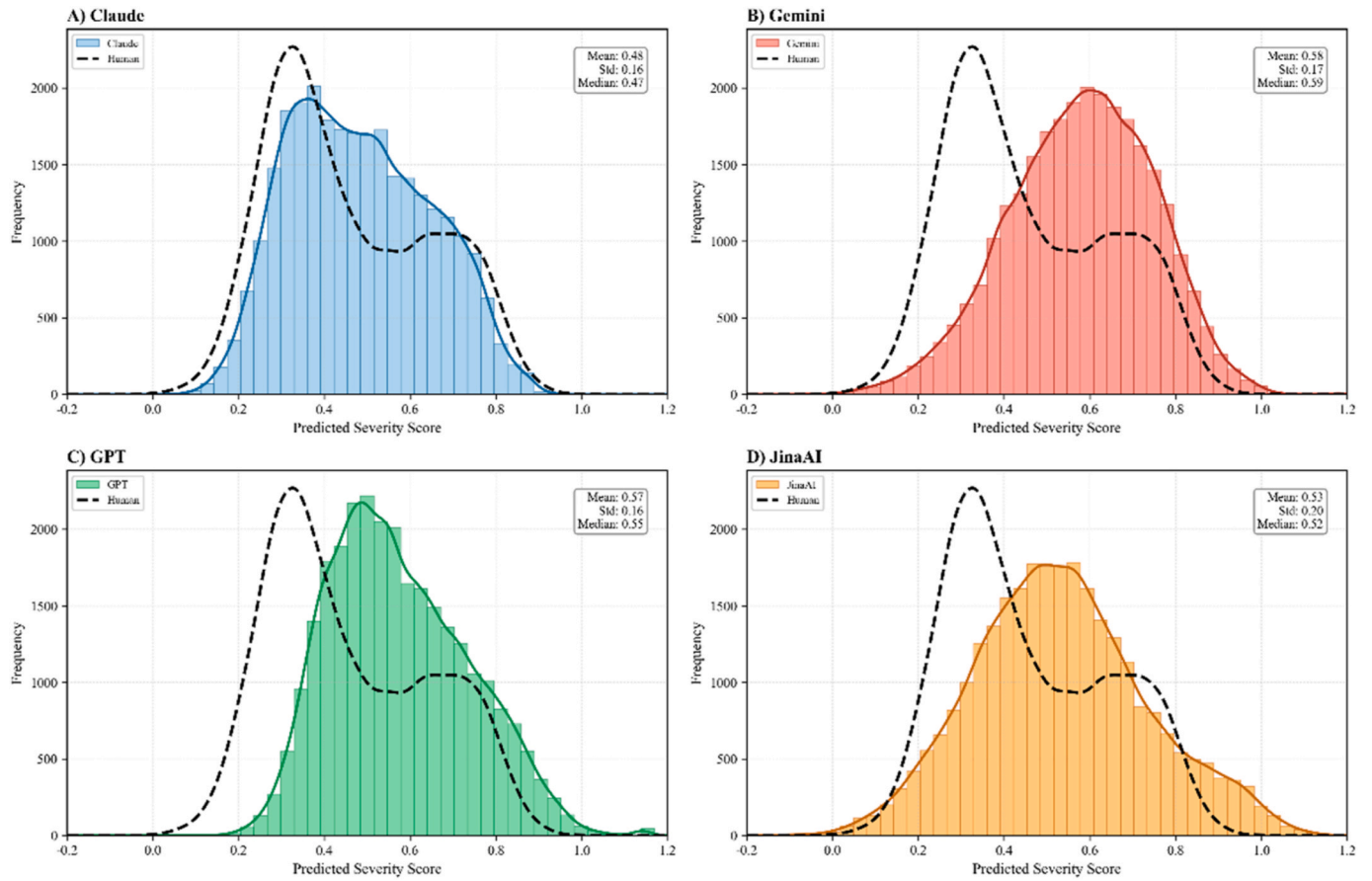


Fig. 4. Distribution of noise complaint predictions based on LLM-Based models.

temporal information (such as “daytime” or “evening”). High-intensity complaints (score >0.6) exhibit the following typical characteristics: explicit high-decibel noise sources (electric drills, heavy machinery, loud speakers), unreasonable occurrence times (late night, early morning, lunch break), long-term persistence (weeks, months), extensive impact range (entire building, entire residential complex), and specific health hazard descriptions (insomnia, anxiety, children’s learning

affected).

To validate the advantages of our method over traditional sentiment analysis, we compared our results with those from [Lu et al. \(2025\)](#) based on sentiment analysis approaches. Their method outputs sentiment polarity scores between 0 and 1, where 0 represents extremely negative and 1 represents extremely positive. We inverted these scores (1 minus the original score) so that higher scores correspond to more severe

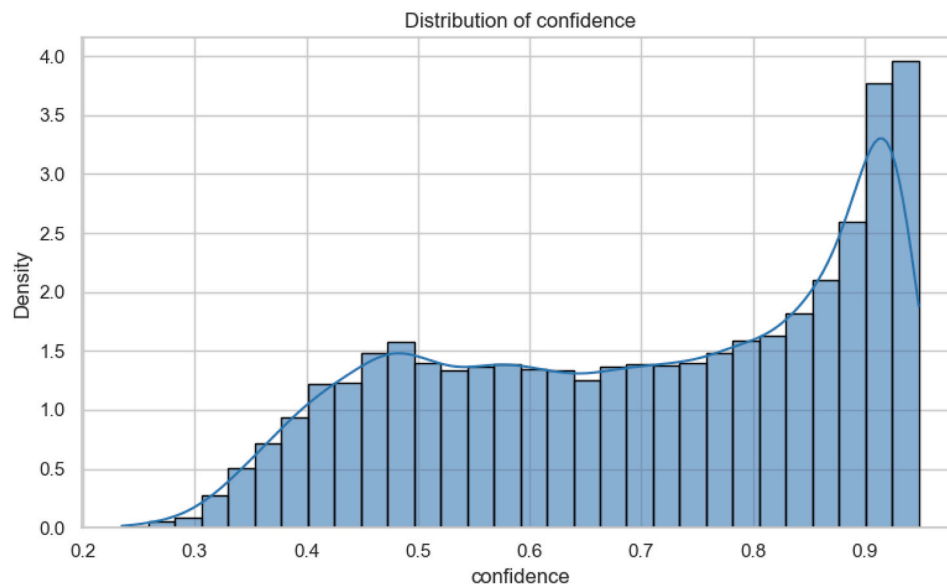


Fig. 5. Distribution of noise complaint predictions based on traditional sentiment analysis model.

complaints (as shown in Fig. 5). Compared to our approach, the traditional method exhibits serious classification problems, with most scores clustering between 0.5 and 0.9. This indicates that without noise exposure data participating in prompting or fine-tuning, traditional sentiment analysis suffers from significant bias, severely compromising result reliability.

3.2. Calculation of the dose-response effect of green space exposure on noise complaint intensity

To examine the relationship between green space exposure and noise complaint intensity, we first constructed environmental buffer zones around each complaint location to extract local exposure variables. As shown in Fig. 6, we compared the Akaike Information Criterion (AIC) values of GAMs across different buffer radii at 10 m intervals. A total of 50 GAMs were fitted, and the 170 m buffer showed the lowest AIC value across the LLM-based complaint intensity measures, indicating favorable model performance at this scale. To reduce potential bias caused by spatial smoothing, we further conducted a scale-stability analysis of GAM partial response curves. Specifically, the NDVI, the tree GVI, the grass GVI, and road density were recalculated from the raw data at each candidate buffer radius, and GAMs were refitted under the same model specification.

We then calculated the standardized distance between adjacent partial response curves. As shown in Appendix Fig. 1, these distances remained generally below the 0.05 reference threshold from 170 m onward, suggesting that further increasing the buffer radius did not substantially change the nonlinear response patterns of the main explanatory variables. Therefore, 170 m was selected as the minimum stable local exposure scale for subsequent analyses.

The GAM model constructed based on 170-m buffers demonstrated a good fit across all 20,547 complaint data points. The model achieved a Normalized Root Mean Square Error (NRMSE) of 11.65%, indicating relatively small prediction errors and good model performance. Regarding different types of green space in Fig. 7, we found that the NDVI and the Grass Ratio exhibit broadly similar decreasing response patterns, (the horizontal axis represents gradually increasing element values, the vertical axis represents partial effects, the shaded bands around curves represent 95% confidence intervals, and the horizontal distribution at the bottom indicates the density of complaint points at these values), but this similarity should be interpreted as partial overlap in their greenness-related information rather than as strong collinearity.

Specifically, when exposure to green spaces with a smaller three-dimensional volume (such as grassland) is relatively low, not only does it fail to provide mitigation effects, but it may even promote noise complaint intensity. In contrast, the tree ratio shows a more complex nonlinear pattern and does not indicate a simple monotonic mitigation

effect, suggesting that the role of vertically structured greenery may be context-dependent and may not be fully captured by the visual tree ratio alone (the NDVI threshold approximately 0.28, SVI pixel ratio, grass threshold approximately 0.02, SVI pixel ratio).

However, as tree coverage increases (based on GAM regression calculations), the mitigation effect on complaint intensity actually decreases (threshold approximately 0.12). Our control variable, road density, also exhibits positive and negative trend changes in its impact on noise intensity (threshold approximately 0.01), initially showing suppressive effects followed by amplification effects, opposite to the pattern of low-lying vegetation. These results suggest that the effects of green space and road density on complaint intensity are not linear, and that threshold ranges should be considered in urban greening and noise mitigation planning. Additionally, we found that similar response patterns of the NDVI and the Grass Ratio suggest partial overlap between satellite-derived greenness and low-level visible greenery, but the correlation analysis indicates that this overlap does not constitute severe collinearity.

In addition, we assessed potential multicollinearity among the explanatory variables using Spearman correlation coefficients and VIF values (Appendix Fig. 2). The VIF results indicated no severe multicollinearity among the variables included in the GAM. Some pairwise correlations were statistically significant; however, their effect sizes were very small, such as the correlations between complaint intensity and NDVI ($r = 0.03$), tree GVI ($r = 0.04$), and road density ($r = -0.04$).

The correlation between the grass GVI and the tree GVI was weak, indicating that they capture different dimensions of visible greenery (Appendix Fig. 2). Specifically, the NDVI mainly reflects vegetation greenness and vegetation vigor from a remote-sensing perspective, whereas the GVI measures the amount of greenery visible from the street-view perspective. Furthermore, the supplementary machine learning analyses also supported the rationality of our variable selection. As shown in Appendix Figs. 3 and 4, the Random Forest and XGBoost models used the NDVI, the tree GVI, the grass GVI, and road density within the 170 m buffer to predict noise complaint intensity. The results showed that these variables provided useful predictive information for noise complaint intensity, suggesting that the selected environmental indicators are relevant to explaining variations in complaint intensity.

4. Discussion and conclusion

This study systematically evaluated the nonlinear mitigation effects of urban green spaces on noise complaint intensity by integrating large language models, satellite remote sensing, street view imagery, and generalized additive models. The research confirms that complaint intensity quantification methods based on large language models (including Claude 3.5 Sonnet, GPT-4o, Gemini 2.0 Flash, and Jina

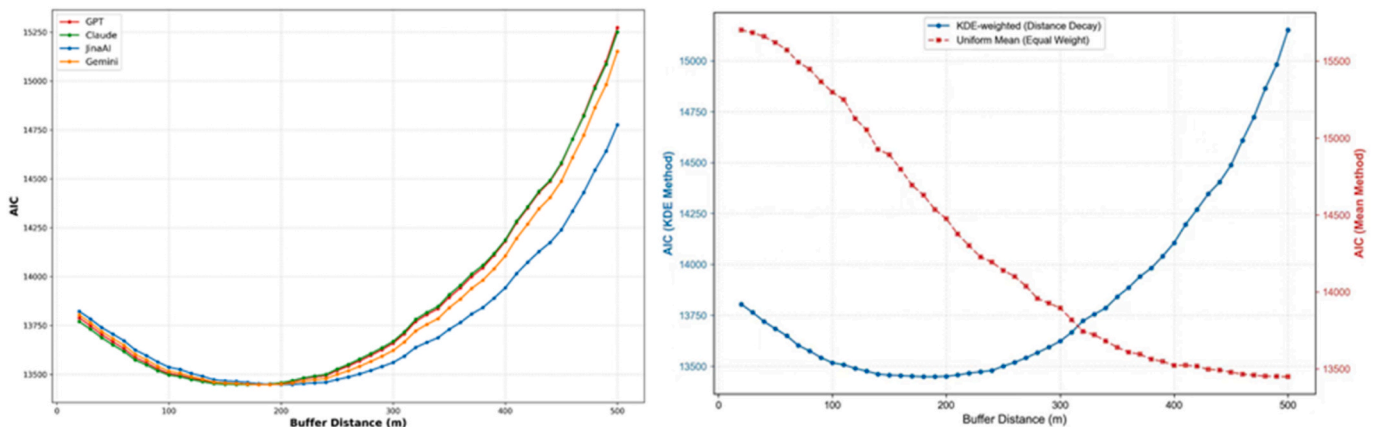


Fig. 6. Buffer distance selection based on AIC comparison.

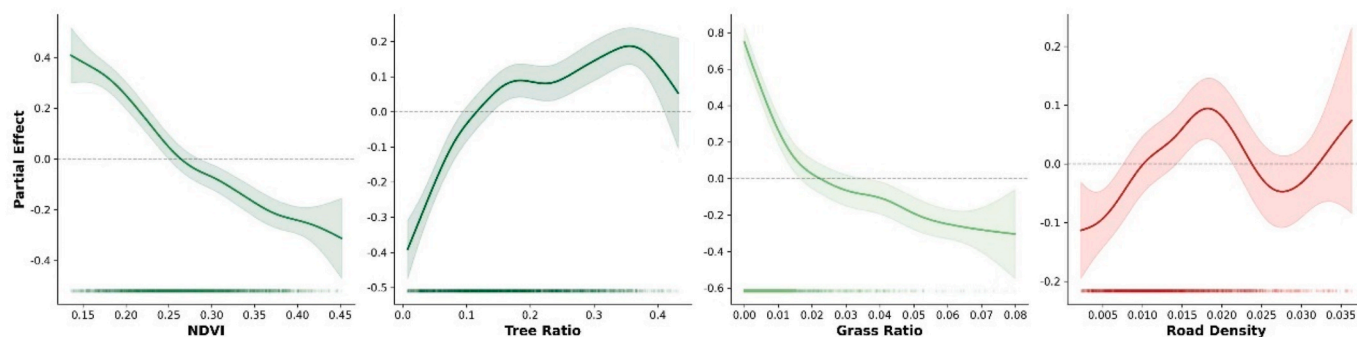


Fig. 7. Non-linear effects of urban greenery exposure and road density on noise complaint intensity: A partial dependence analysis based on GAM

Reranker V2) can achieve more nuanced complaint intensity distributions, which are more conducive to identifying strongly negative complaint content from vulnerable populations. This approach significantly outperforms traditional sentiment analysis. The fine-tuned RoBERTa regression model driven by large language models successfully transferred ranking knowledge to 20,547 complaint data points while maintaining a high accuracy of 61.48%. Additionally, we accurately captured spatial scale dependencies and dose-response thresholds.

The findings indicate that different types of green space exhibit distinct associations with complaint intensity because they represent different dimensions of vegetation exposure. The NDVI captures overall greenness from a remote-sensing perspective, whereas the GVI reflects pedestrian-level visible greenery, and the grass and tree GVI further distinguish vertical vegetation structures. Trees and vertically structured greenery show stronger associations with reduced complaint intensity than low-level grass coverage, likely due to combined visual shielding, partial acoustic attenuation, and psychological restoration effects.

The results emphasize that urban greening should shift from quantitative expansion to qualitative optimization, prioritizing three-dimensional and vertical greening in high-density urban areas, especially the planting of tall trees rather than flat lawns, as the former demonstrates more significant effects in alleviating noise complaints. Green space planning should focus on critical dosage thresholds, such as an NDVI threshold of approximately 0.28, a grass threshold of approximately 0.02 (SVI pixel ratio), and a tree threshold of approximately 0.12 (SVI pixel ratio). Landscape design within optimal dosage ranges can better protect noise-vulnerable and noise-sensitive populations (Korpilo et al., 2024).

This study provides methodological innovation and empirical evidence for intelligent analysis of noise complaints and nature-based noise mitigation strategies. Benefiting from the broad transferability and powerful comprehension capabilities of large language model technology, the methodological framework can be extended to intelligent analysis of other environmental complaints such as air pollution and waste management. This establishes a technical foundation for constructing comprehensive smart city environmental complaint management platforms, advancing environmental justice and sustainable urban planning and development. Because the complaint data lack reliable individual-level demographic information, this study cannot examine differences in perceived noise intensity across population groups. Future

research integrating socioeconomic, traffic, and fine-grained land-use data should further clarify the mechanisms linking green space exposure and perceived environmental disturbance, supporting environmental justice and sustainable urban planning.

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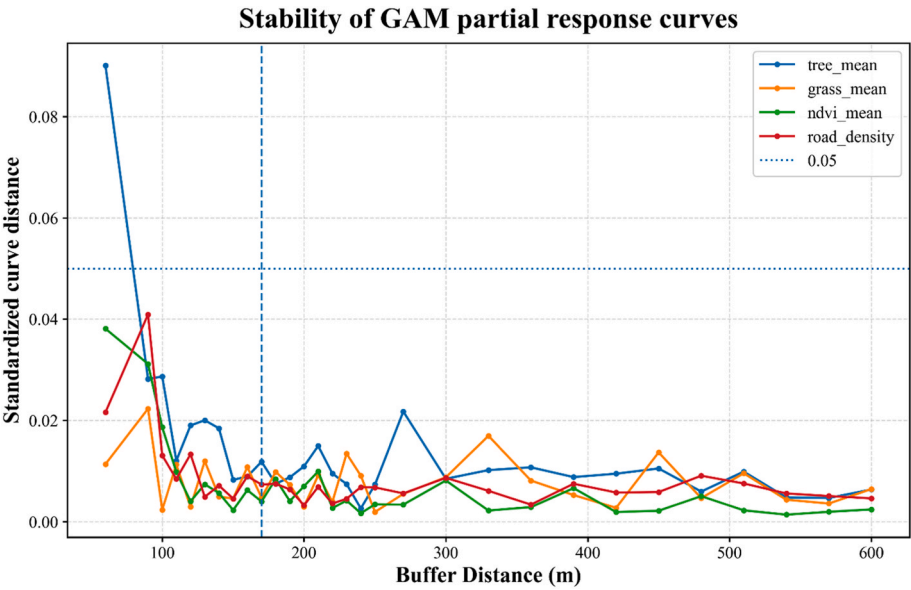
CRediT authorship contribution statement

Yan Zhang: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Zhijie Zhang:** Data curation, Software, Visualization. **Mei-Po Kwan:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing. **Wei Wang:** Data curation, Writing – review & editing.

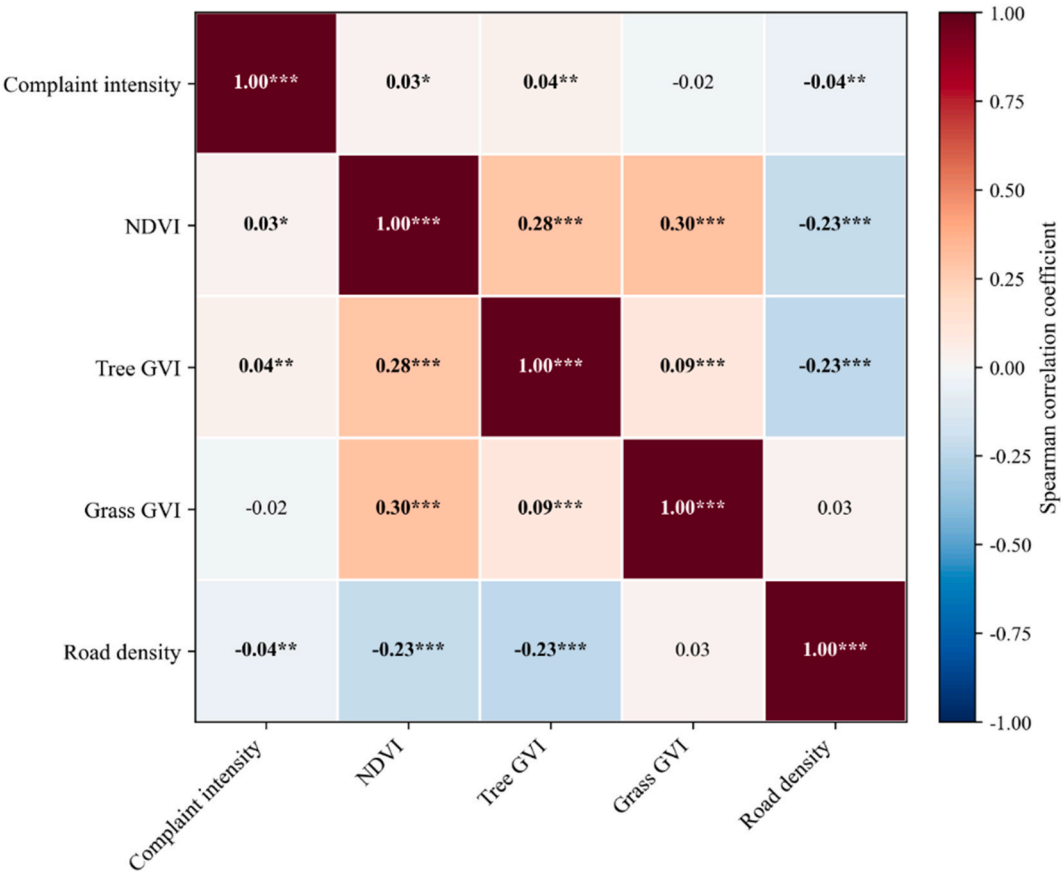
Declaration of competing interest

The authors declare that they have no financial and personal relationships with other people or organizations that can inappropriately influence our work.

Appendix

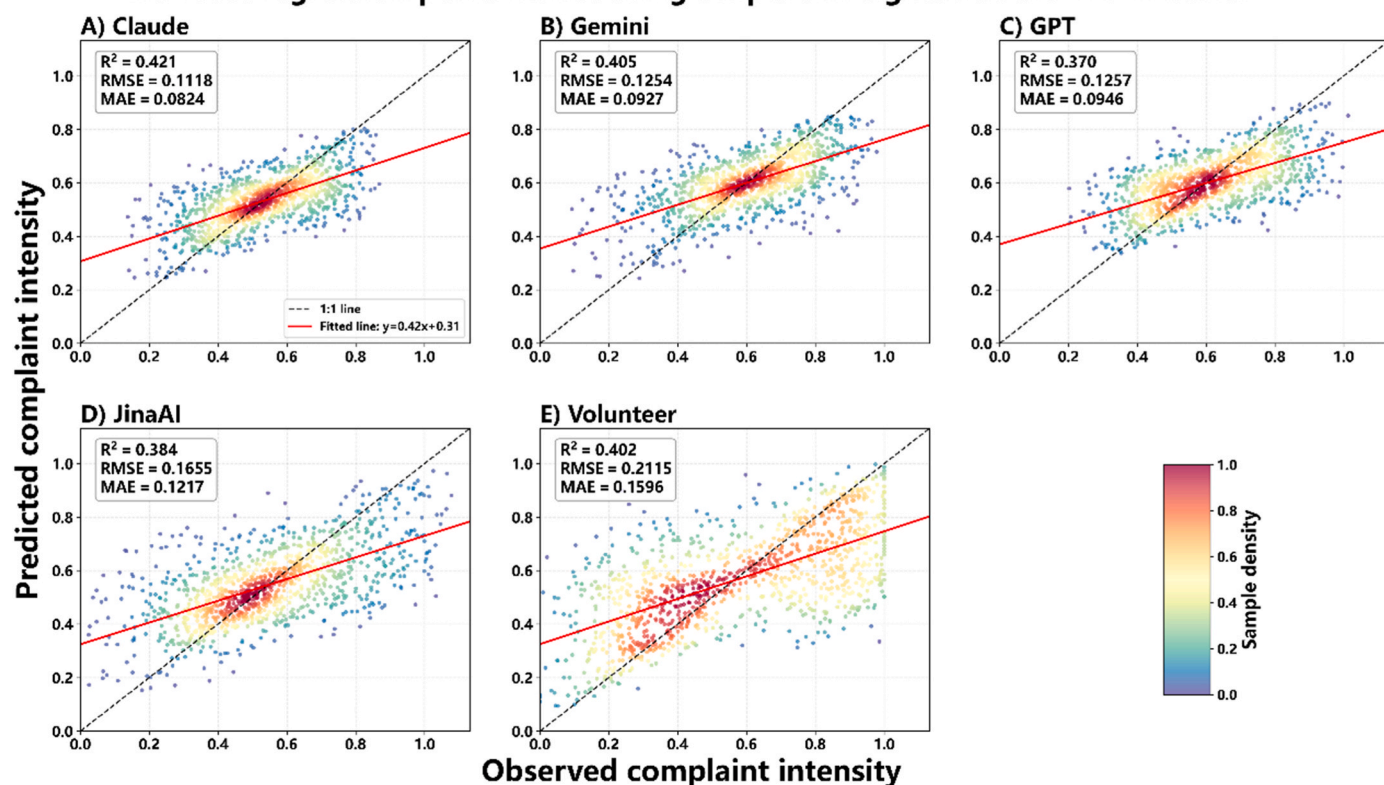


Appendix Fig. 1. Standardized distances between adjacent GAM partial response curves were calculated for the NDVI, the tree GVI, the grass GVI, and road density. The horizontal dotted line indicates the 0.05 reference threshold, and the vertical dashed line indicates the selected 170 m buffer. From 170 m onward, the curve distances remain generally below the threshold, indicating stable nonlinear response patterns across larger buffer scales.



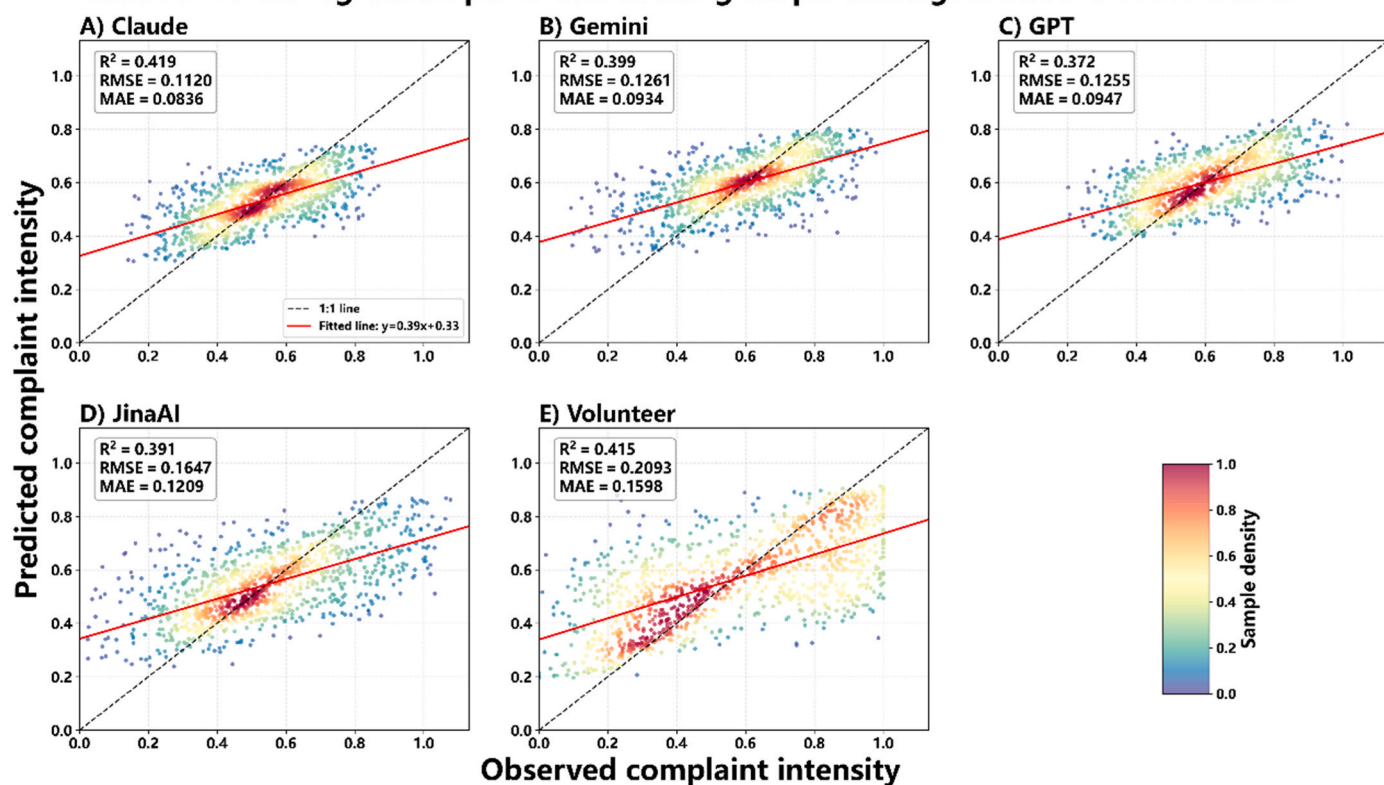
Appendix Fig. 2. The correlation figure: (the correlation between complaint intensity and the NDVI is only $r=0.03$, the correlation between complaint intensity and the tree GVI is $r=0.04$, and the correlation between complaint intensity and road density is $r=-0.04$. These results indicate that the statistically significant correlations should be interpreted with caution, as their effect sizes are negligible.).

XGBoost regression performance using simple unweighted at the 170 m buffer

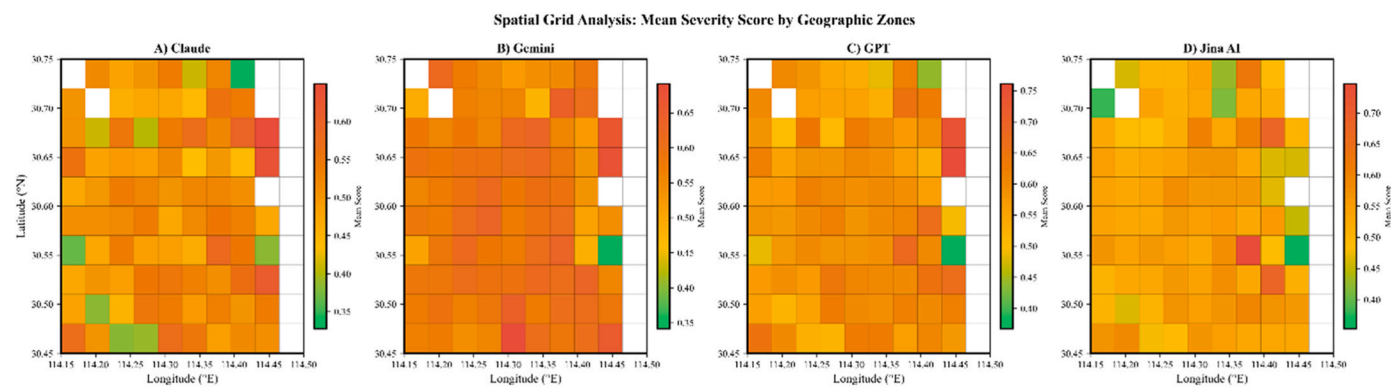


Appendix Fig. 3.

Random Forest regression performance using simple unweighted at the 170 m buffer



Appendix Fig. 4.



Appendix Fig. 5. Spatial distribution of mean noise complaint severity scores inferred by four LLMs across geographic grid cells. Warmer colors indicate higher mean severity scores, while cooler colors indicate lower mean severity scores.

Data availability

The data that has been used is confidential.

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